Advanced Computer Applications

ASSIGNMENT 2

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Parametric Sketching

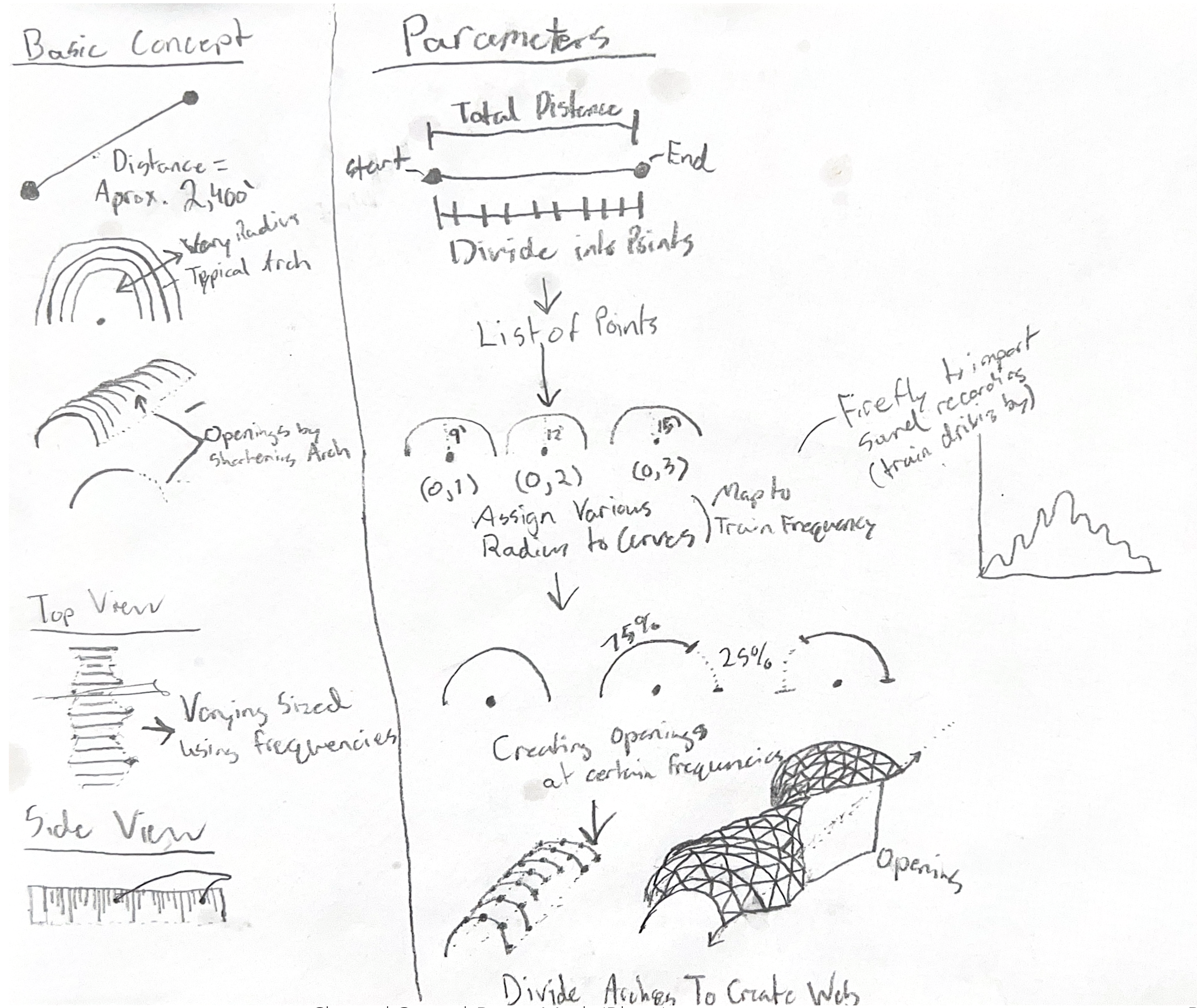
Project Description

This skecth started with the concept of a arched form that creates a semi shaded path. I wanted these arches to be varying sizes and additionally peel up at varrying points to create openings in the side.

This concept bases the idea of having a linear path that could be different lengths and then this arched form would be be stretched across the distance..

As that path gets divided a variety of radius's get assigned to the points to create the undulating form. This then goes one step further and the end points get remapped to different heights to create the openings.

The structure of this is based on a triangulation which then is the outline for a piped structure.



Cleaned Ground-Points via Cloud Compare

Cleaned Ground-Points Mesh (Rhino)

Revit Topo-Solid

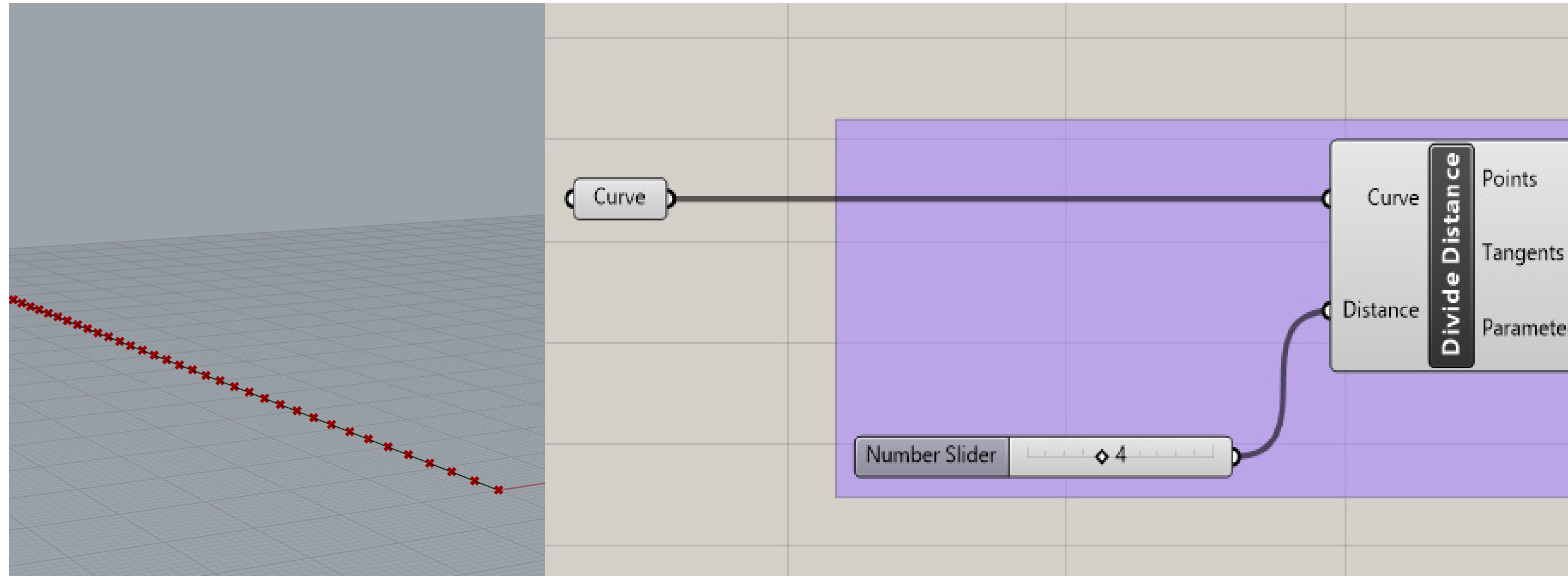
Modeling Rhino & Grasshopper

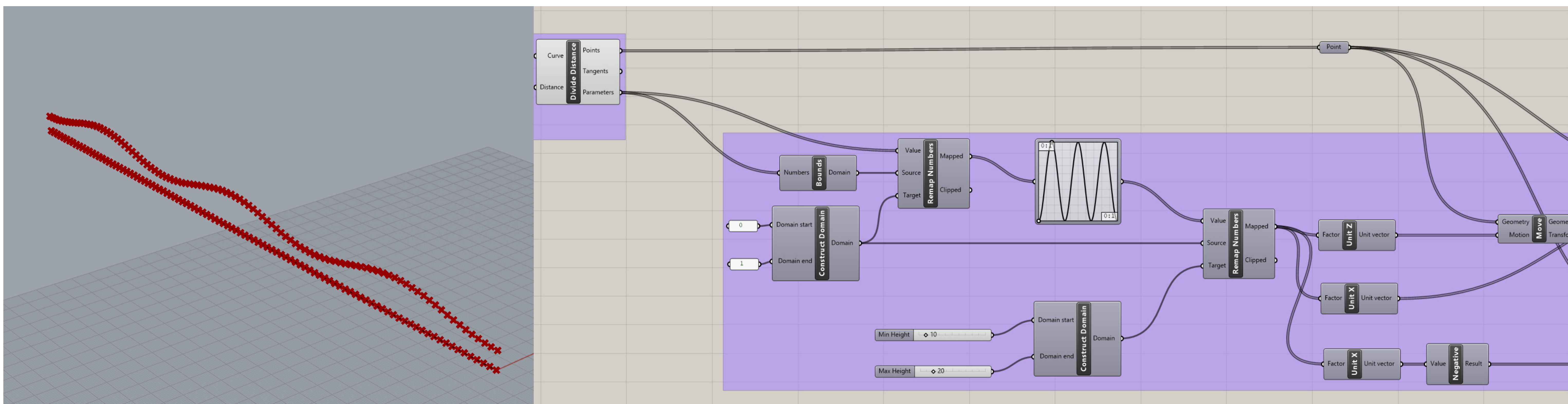
Script Breakdown

This is the start of the cript that requires a manual input of a curve which is what defines the length and shape of the path.

It begins by dividing the curve which can be adjusted in order to gain more detail and control points.

The points that are an output of dividing the curve is what gets remapped. The ramapping is based on a sin grapgh, assigning defined values from the domain. This part can be changed to adjust the minnimum and maximum heights/radius.





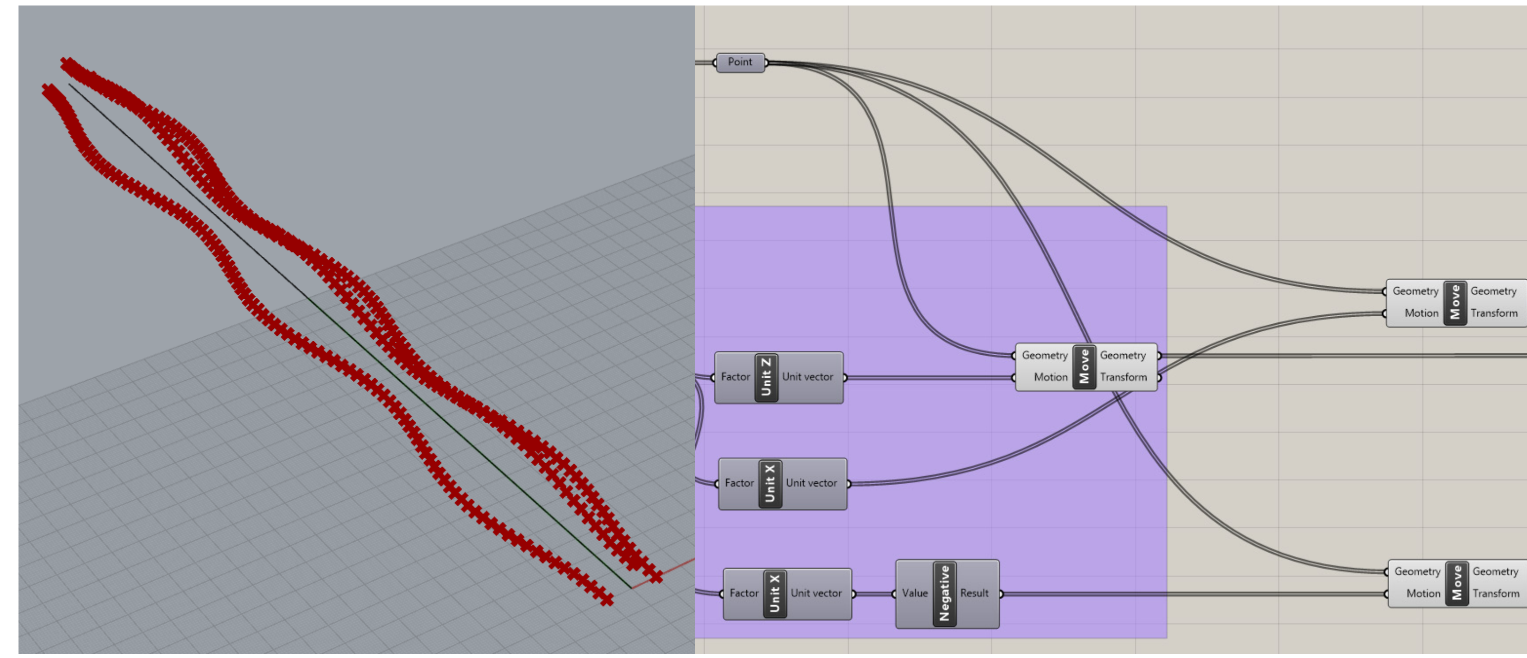
Modeling Rhino & Grasshopper

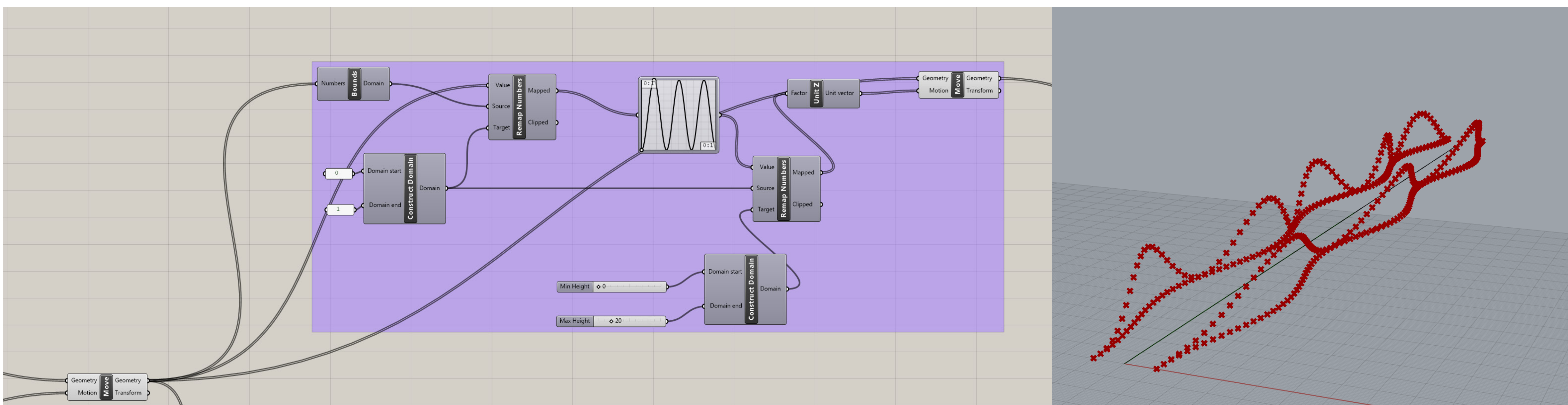
Script Breakdown

This the exact same process as the first remapping but instead of on the Z axis, it is on the X axis. This is in both directs (positive/negative).

The next step is to create the openings which is also based on a sin graph. This is remapping the perimiter points that moved on the X axis. These now get remapped on the Z axis to develop undulating openenings.

Everything that remapped by the graph mapper can be simply adjusted by moving the sliders on the graphs.



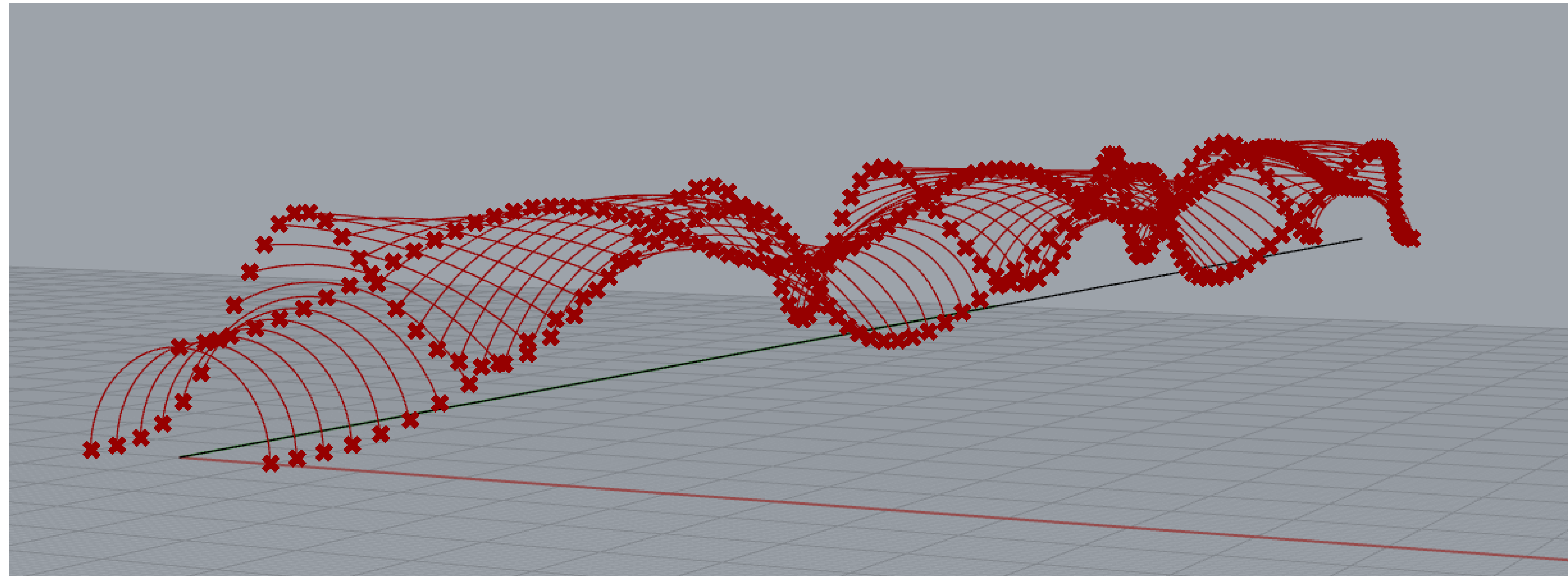


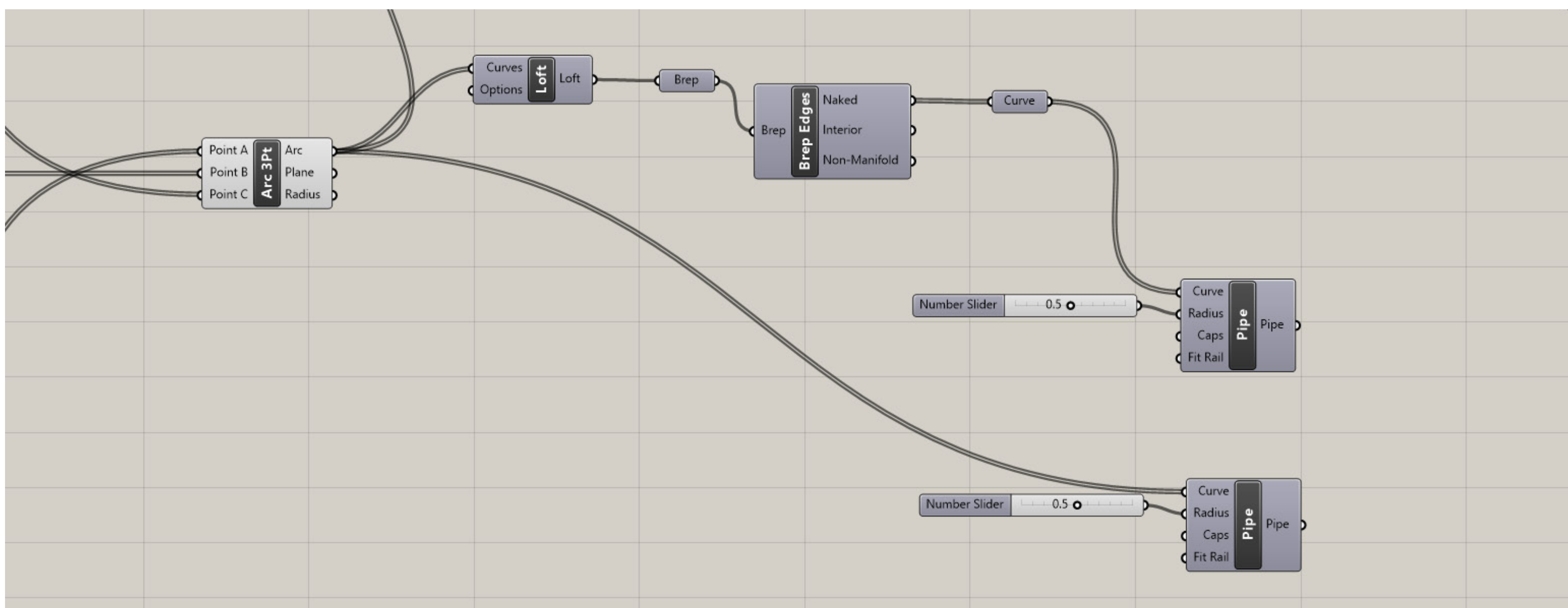
Modeling Rhino & Grasshopper

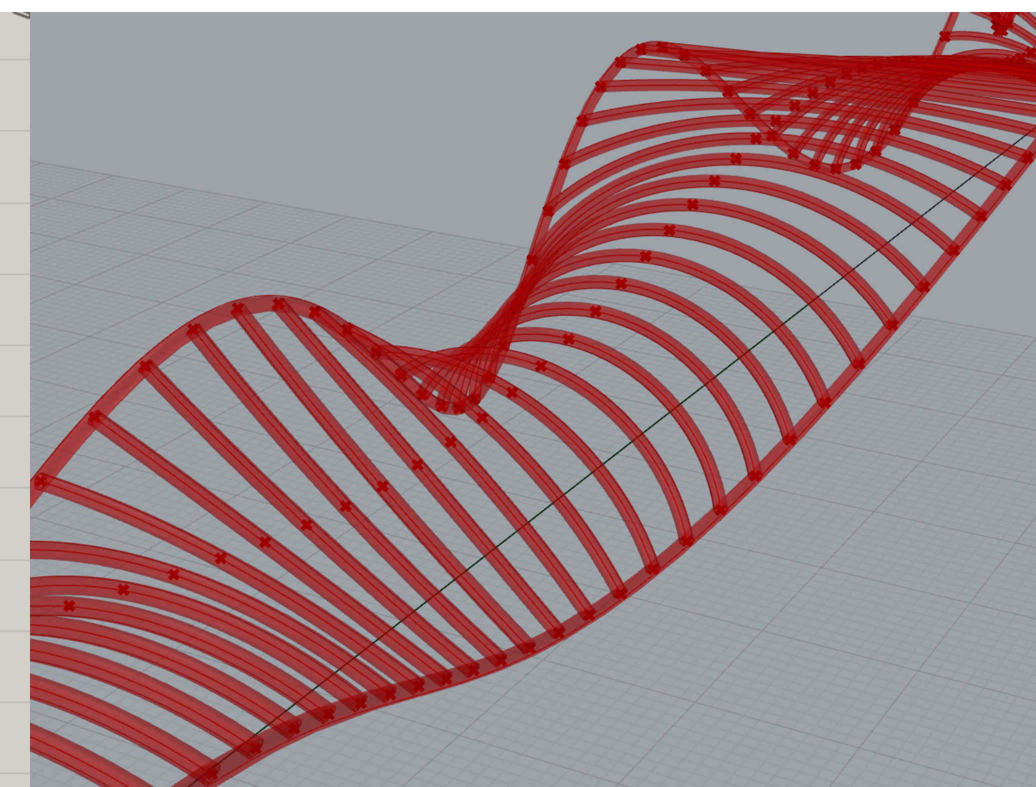
Script Breakdown

Now this is where the arc's are created based on the 3 different set of points. These curves are used to create the main cross beams.

Then a loft command is used to extract a perimeter outline of the structure.





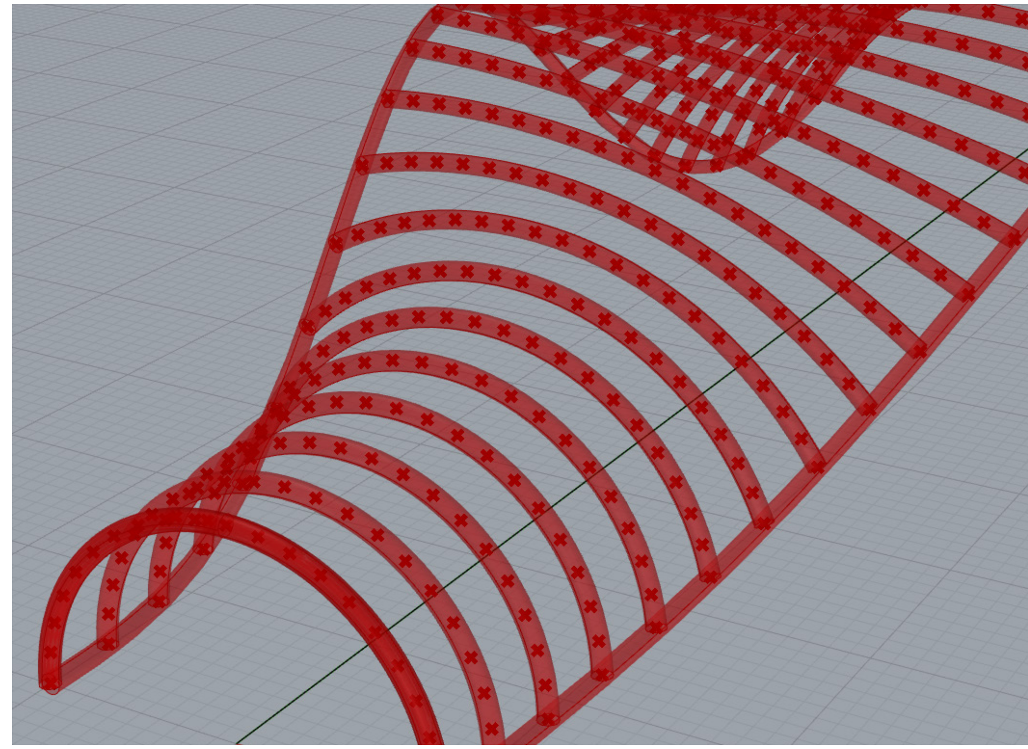


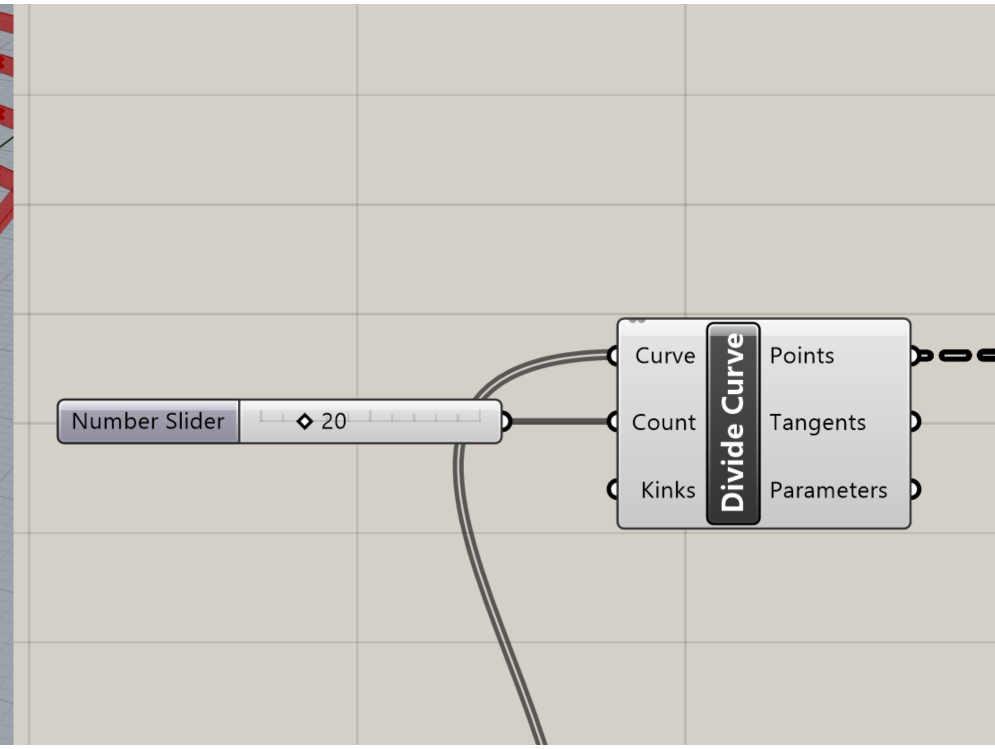
Modeling Rhino & Grasshopper

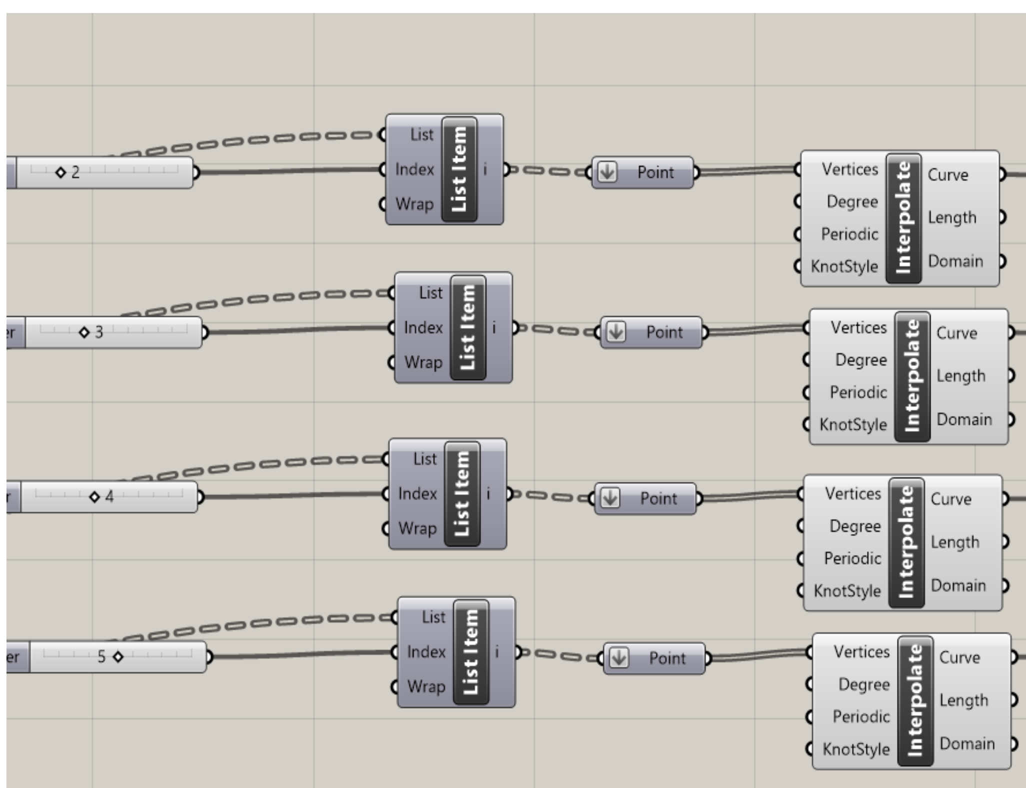
Script Breakdown

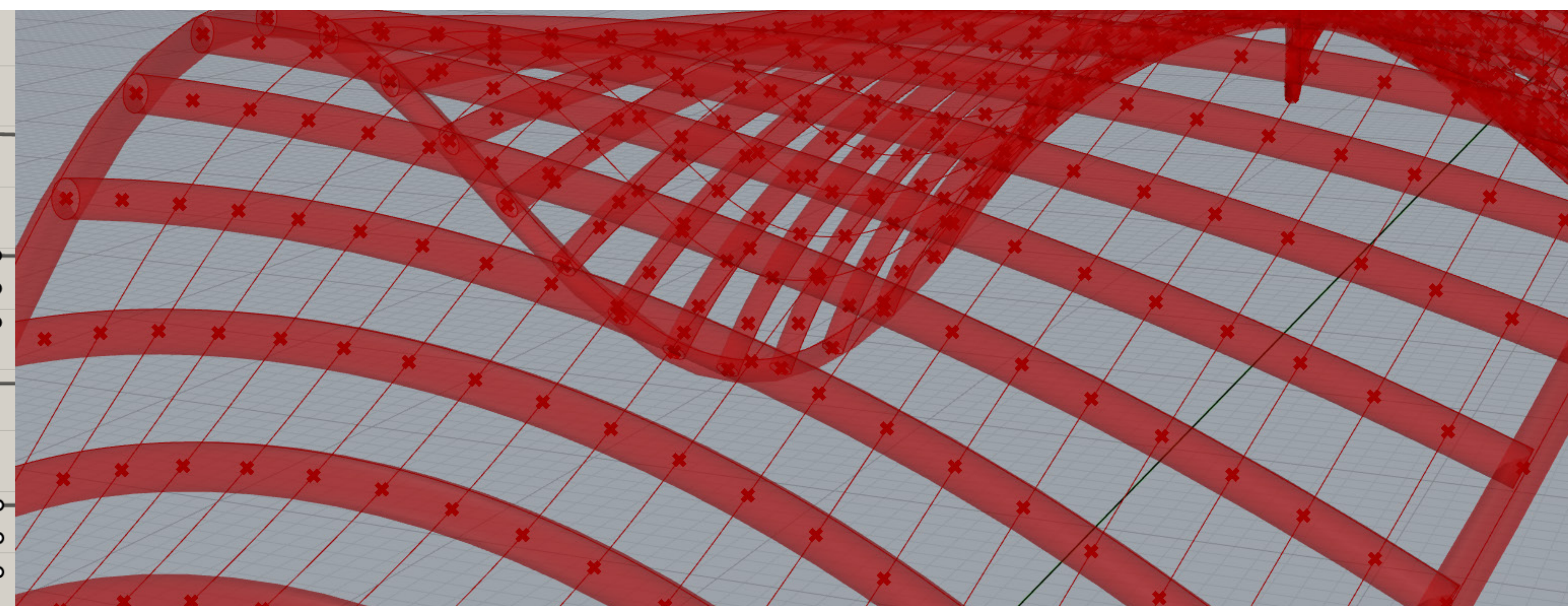
This is a portion of my script that I know can be simplifed but I did not know how to do it.

The main arc's are divided to create a substructure and then a curve is interpolated across all of the points.





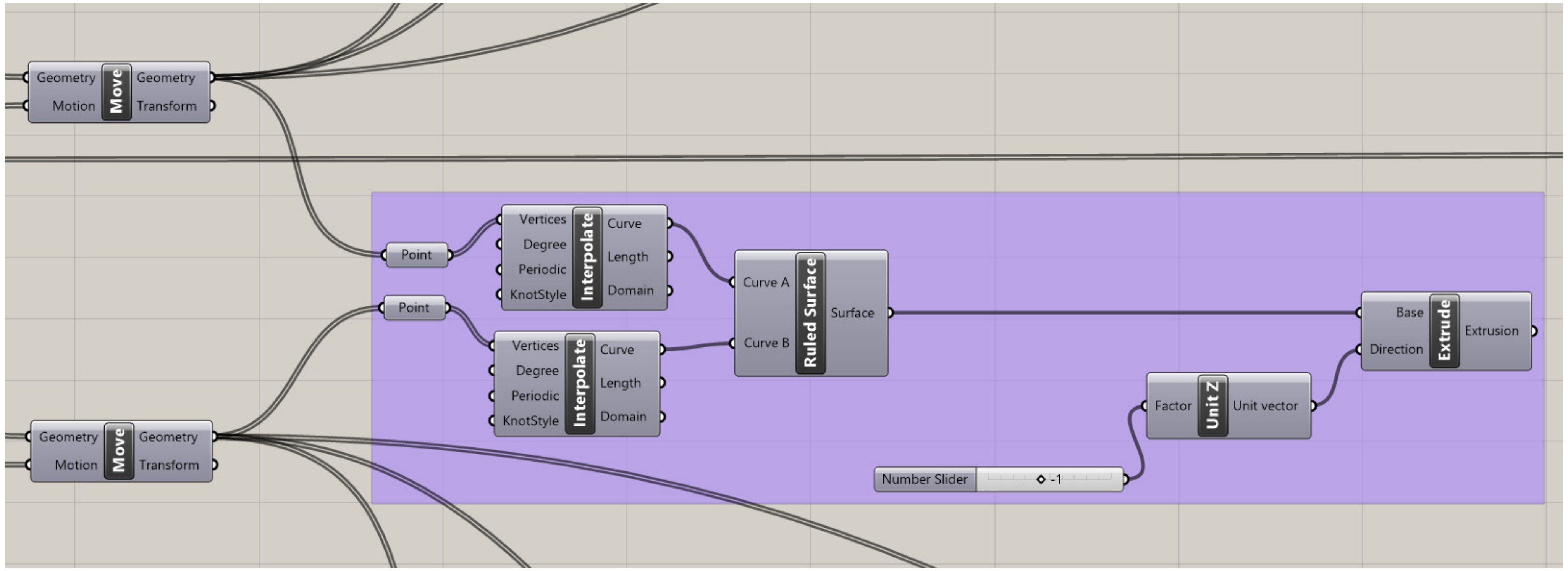


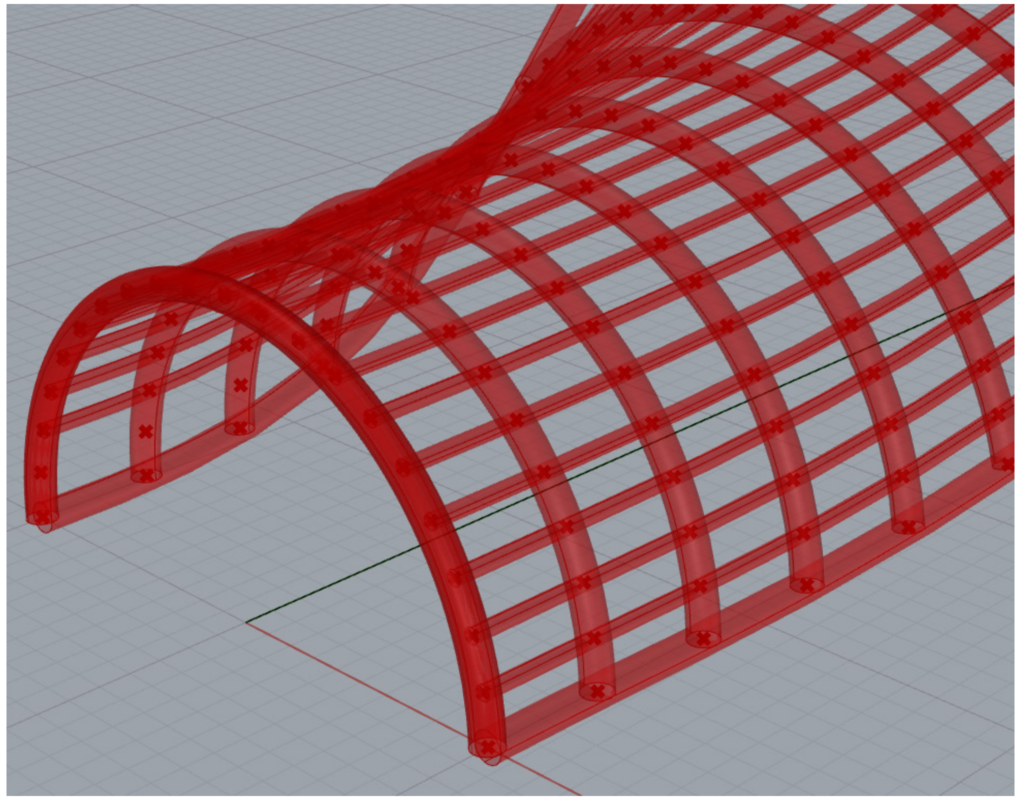


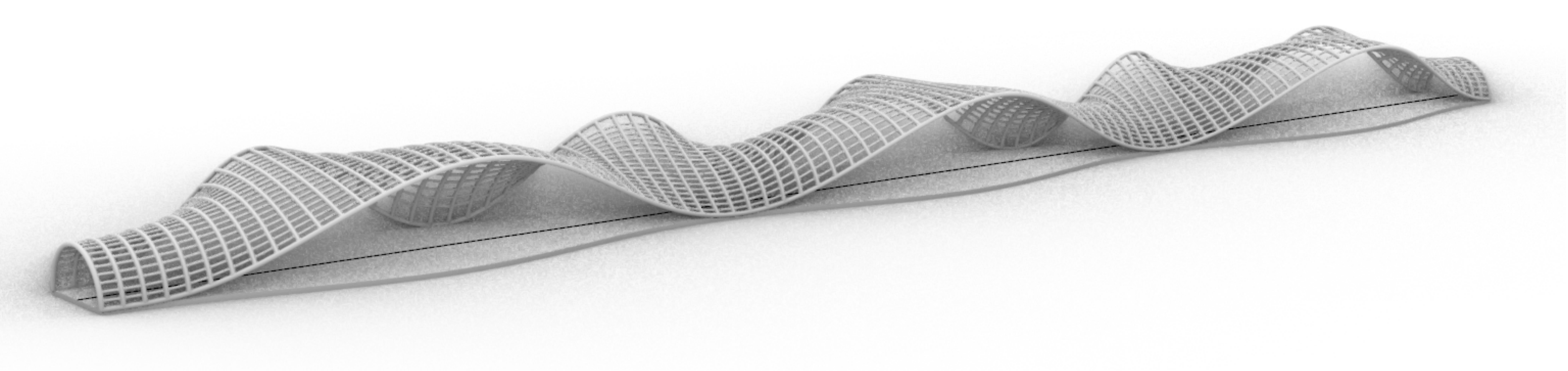
Modeling Rhino & Grasshopper

Script Breakdown

This final step was to create a floor that is based on the outline of the structure. This was done by taking those perimeter points and creating a ruled surface and then extruding.





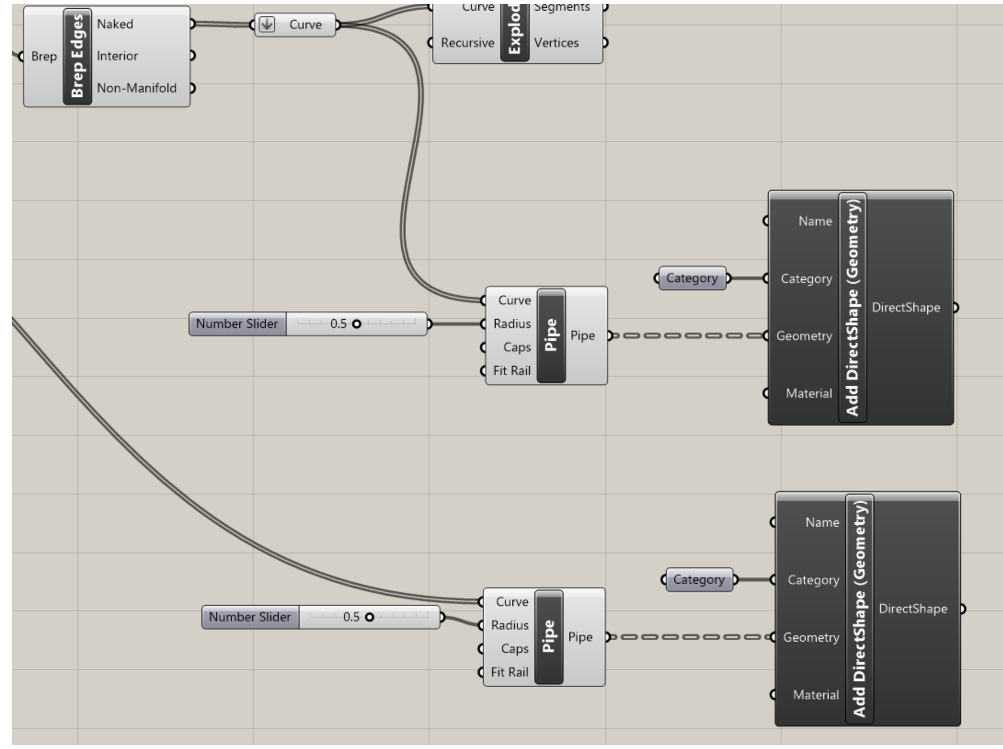


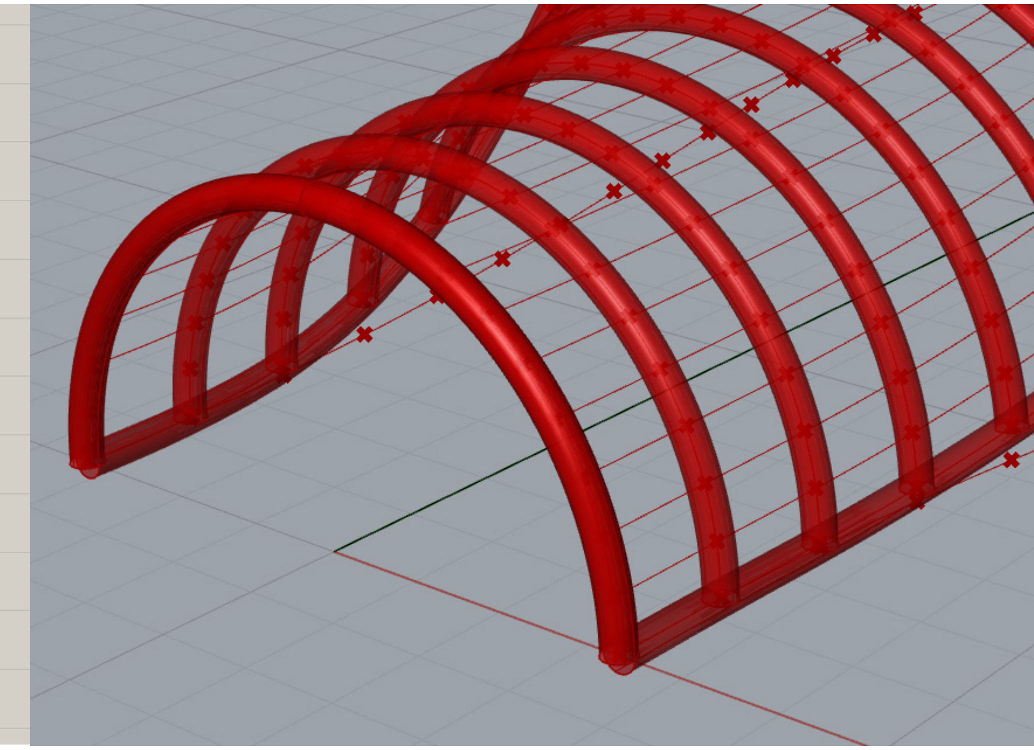
Documentation Rhino-Inside-Revit

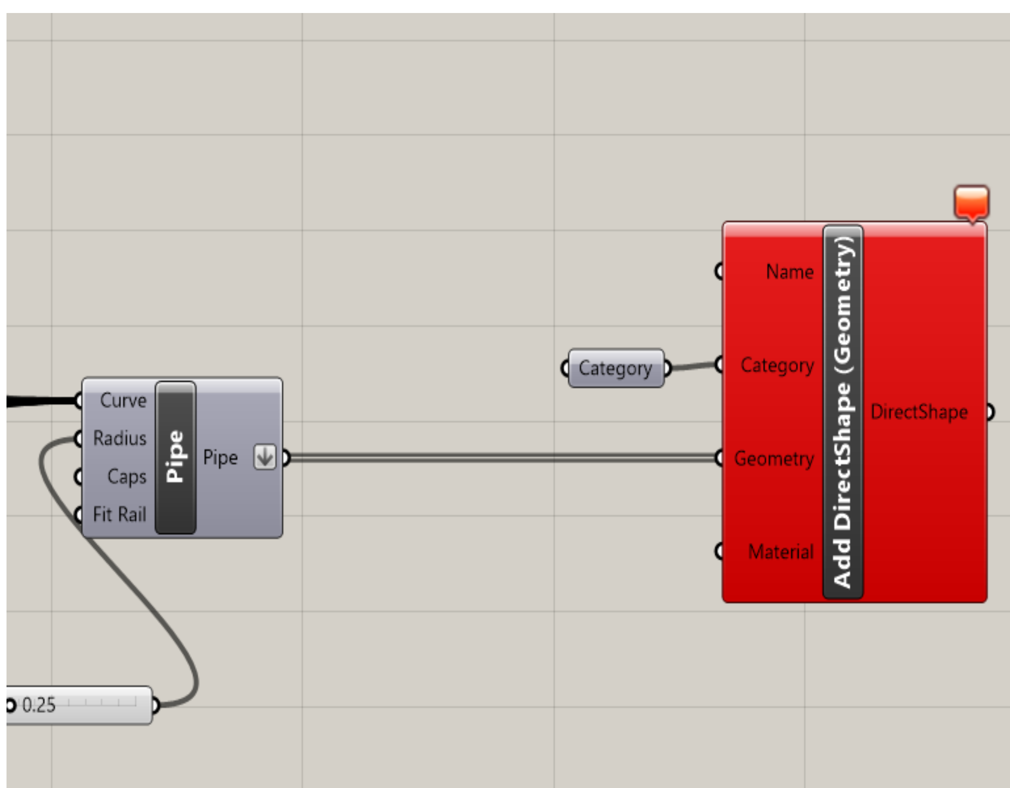
Model Transfer

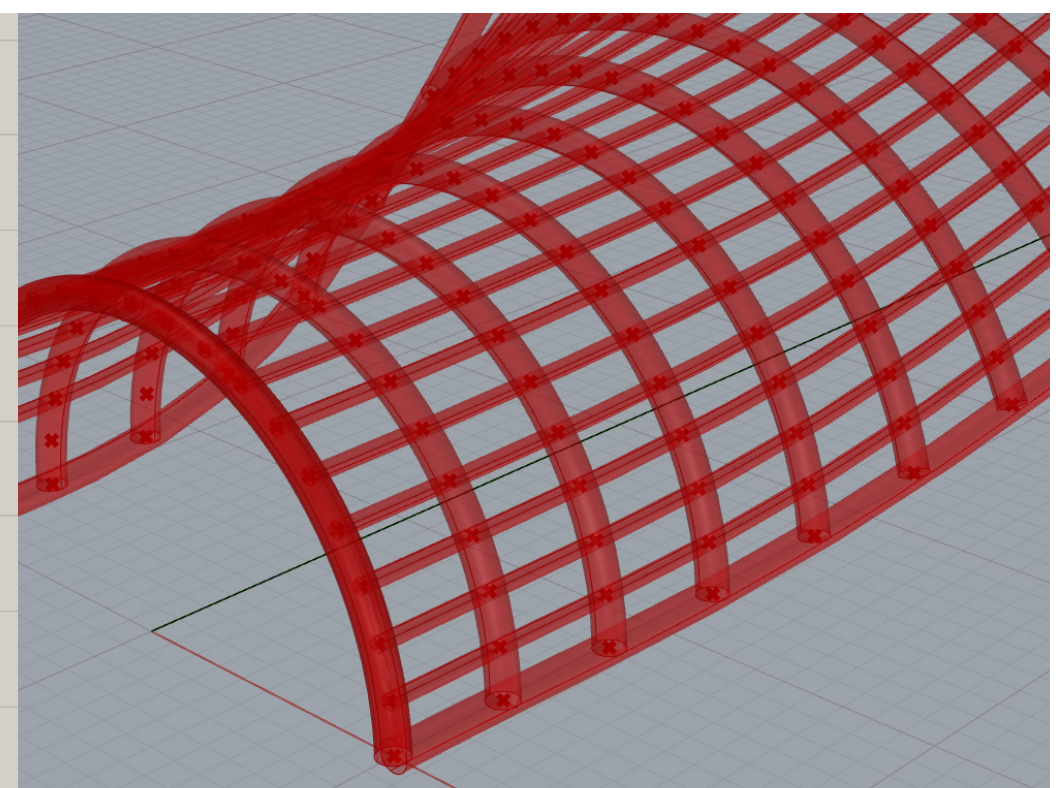
To be able to get this structure into Revit I ended up using a Directshape comand instead of RevitBeam. The Beam command was not fully working and giving errors.

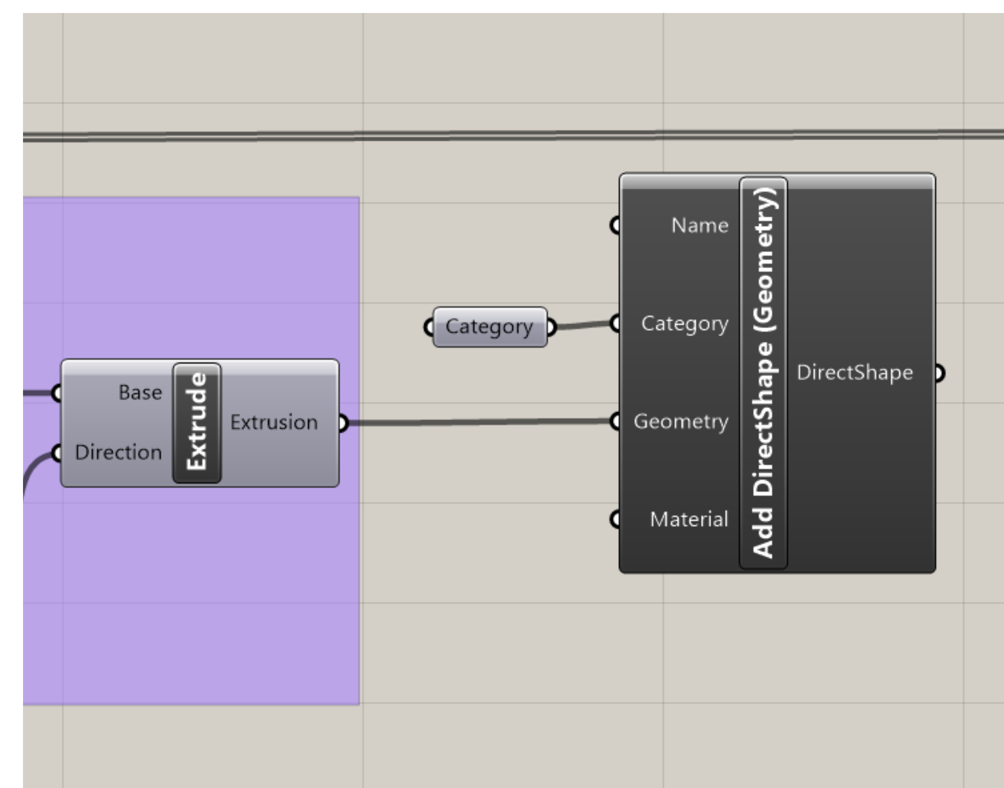
Fortunatley this command still allows you to set the category of the objects so that you can alter them in Revit according to what they actually are.







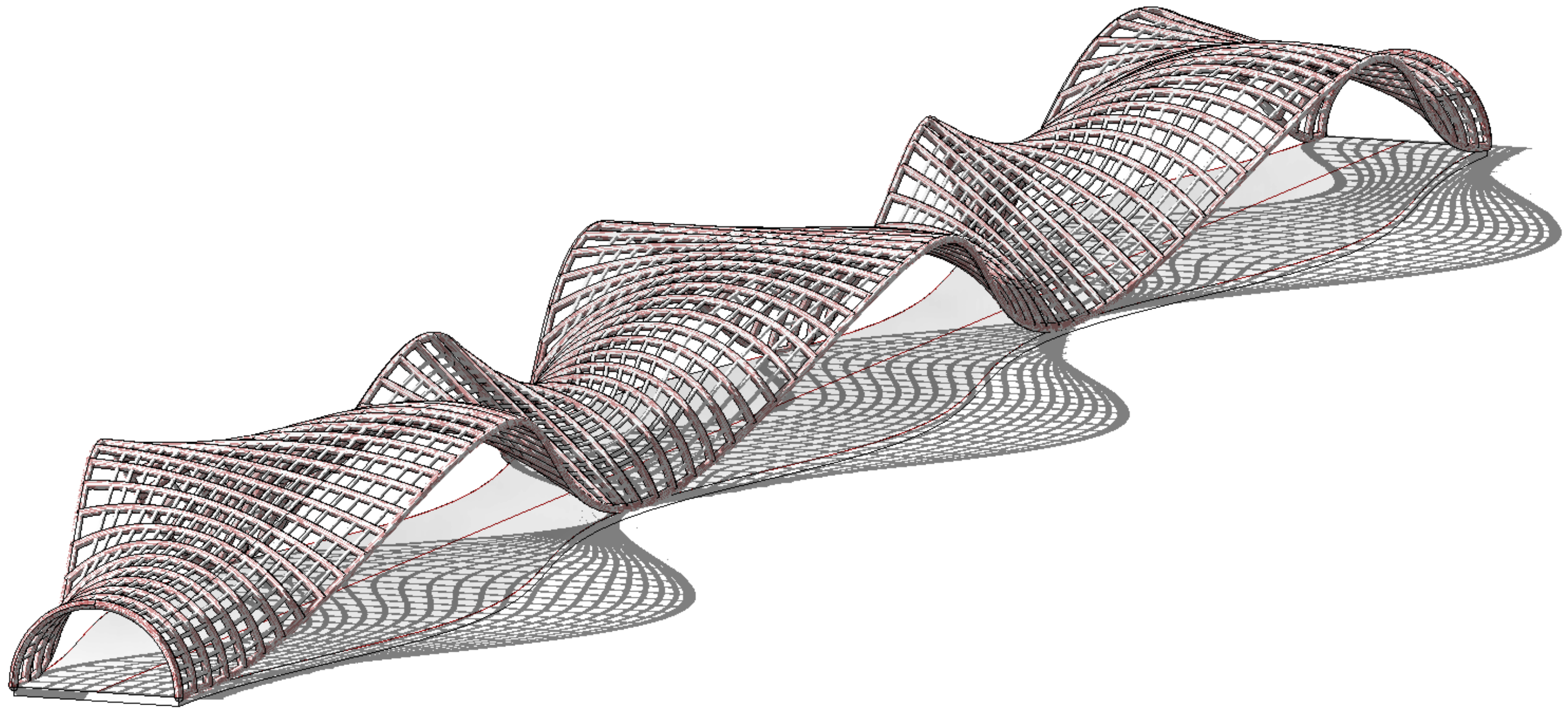


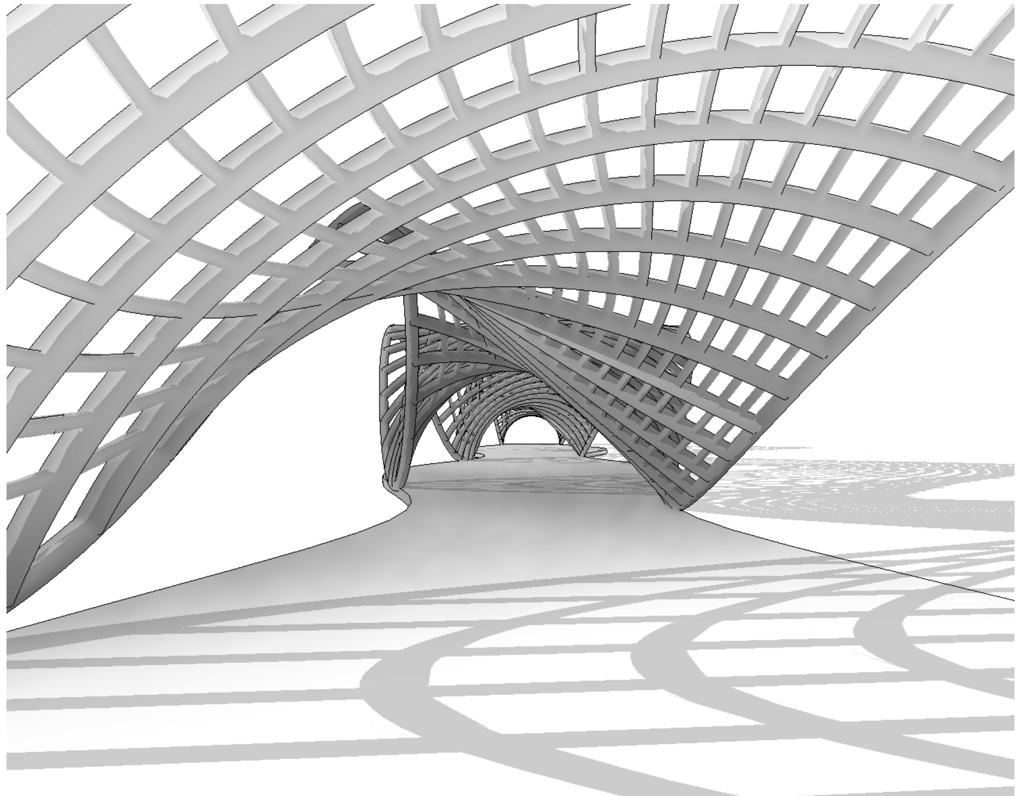


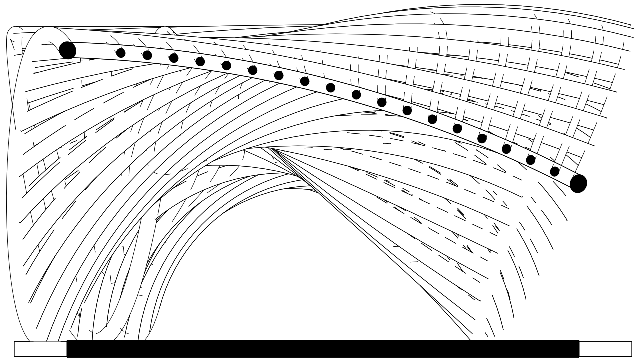
Documentation Rhino-Inside-Revit

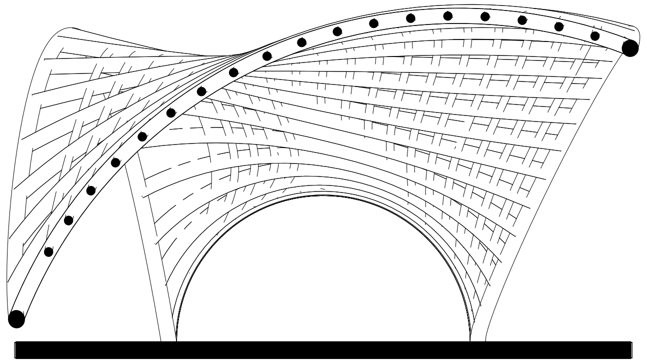
Revit Documentation

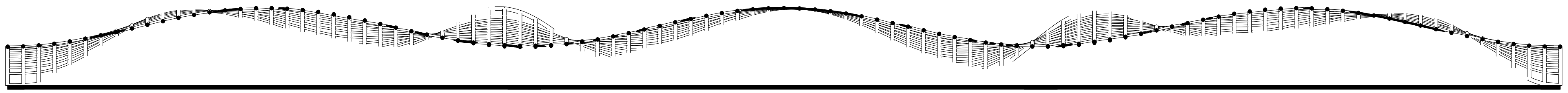
The value of being able to use R-I-R to get a complex Rhino model into Revit is incredible. Revit is ideal for creating drawings super quick and consistent.











Rendering TwinMotion

Desert Oasis

This structure is depicted to be made of steel that has endured the desert climate and conformed to the natural color palette of th landscape. This path is lined with palm trees to give the users a sense of oasis and refreshment in a rather harsh, dry, and uncomfortable setting.